Iteration 4 BDAS Report

A Data Mining Solution for Early Gallstone Prediction
to Support Sustainable Healthcare

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1 Introduction

Sustainable Development Goal 3 (SDG 3), *Good Health and Well-Being* 1, aims to improve health outcomes through prevention, early intervention, and access to effective healthcare services. Achieving this goal increasingly depends on the ability of healthcare systems to use available data efficiently and support evidence-based clinical decisions. Data mining and machine learning can contribute to this objective by identifying hidden patterns in routinely collected health information and supporting earlier detection of disease risk.



Figure 1: 17 Global Challenge for Government Transparency: The Sustainable Development Goals (SDG) 2030 Agenda

Gallstone disease is a common condition that may remain asymptomatic but can lead to complications such as acute cholecystitis, biliary obstruction, cholangitis, and pancreatitis (Fujita et al., 2023; Lammert et al., 2016). Diagnosis generally relies on clinical assessment and abdominal ultrasonography. However, identifying high-risk individuals before complications occur may support earlier investigation and improve the efficient use of healthcare resources.

This project therefore develops a data-mining solution for the early prediction of gallstone disease using demographic, comorbidity, laboratory, and body-composition information. The Gallstone dataset contains records from 319 participants, including both patients with gallstones and control participants (I. Esen et al., 2025; İ. Esen et al., 2024). Machine-learning models will be used to identify patterns associated with gallstone status and estimate individual disease risk.

The proposed solution is intended as a clinical decision-support tool rather than a replacement for medical diagnosis. By evaluating predictive performance, sensitivity, interpretability, and important risk factors, the project aims to demonstrate how data-driven healthcare can support preventive intervention, reduce avoidable complications, and improve resource allocation. In this way, the project contributes to a more proactive, efficient, and sustainable healthcare system aligned with the broader objectives of SDG 3.

2 Business and Situation Understanding

2.1 Clinical background and motivation

Gallstone disease (cholelithiasis) is a common hepatobiliary disorder in which solid deposits form within the gallbladder or biliary tract. Although many gallstones remain clinically silent, obstruction of the cystic or common bile duct can lead to biliary colic and complications such as acute cholecystitis, obstructive jaundice,

cholangitis, and gallstone pancreatitis Fujita et al., 2023; Lammert et al., 2016. A recent systematic review and meta-analysis estimated that approximately 6% of the global population has gallstones, with prevalence varying substantially by sex and geographical region Wang et al., 2024. The scale of the condition and the possibility of clinically important complications make the identification of individuals at increased risk a relevant preventive and diagnostic-support problem.

Gallstone formation is multifactorial rather than the result of a single abnormal measurement. Established or commonly reported associations include age, sex, obesity and body-fat distribution, metabolic abnormalities, dyslipidaemia, and diabetes Lammert et al., 2016. These relationships are relevant to the present dataset because its predictors combine demographic information, comorbidities, body-composition measurements, and routine laboratory markers. This creates an opportunity to investigate whether the joint pattern of routinely obtainable variables contains sufficient information to distinguish patients with and without gallstones.

In current clinical practice, abdominal ultrasonography remains a principal first-line imaging method for suspected gallstone disease, together with appropriate laboratory assessment European Association for the Study of the Liver (EASL), 2016; Fujita et al., 2023; Gutt et al., 2020; National Institute for Health and Care Excellence, 2014. Therefore, the purpose of the proposed machine learning system is *not* to replace ultrasound or provide an autonomous diagnosis. Instead, the project treats prediction as a risk-stratification and decision-support problem: a model could estimate the probability of gallstone disease from non-imaging clinical information and thereby identify individuals who may warrant closer clinical assessment or confirmatory imaging.

This direction is supported by recent work applying machine learning to clinical and metabolic information. Salem et al. combined metabolomic and clinical predictors and demonstrated that nonlinear models can capture patterns associated with gallstone status Salem et al., 2023. More directly, Esen et al. introduced the dataset used in this project and reported that models based on bioimpedance and laboratory variables could distinguish gallstone cases from controls, with their strongest models reaching approximately 85% accuracy and an area under the receiver-operating-characteristic curve (ROC–AUC) of about 0.85 İ. Esen et al., 2024. These results provide a useful benchmark, but they do not remove the need for careful validation, because the available sample is small and originates from a single clinical setting.

2.2 Business and situation objectives

The practical objective of this project is to develop and evaluate an interpretable early-risk prediction approach for gallstone disease using information that may already be available from routine clinical assessment, blood testing, and bioimpedance measurement. In a realistic healthcare workflow, the intended output would be a *risk estimate* rather than a definitive diagnostic label. The model should help prioritise patients for further assessment while maintaining sufficient transparency for its predictions to be examined by a clinician.

Accordingly, the business/situation objectives are:

1. Determine whether demographic, comorbidity, laboratory, and body-composition variables can provide useful early discrimination between patients with and without gallstones;
2. Minimise clinically undesirable false-negative predictions, because a model that frequently classifies gallstone cases as low risk would have limited screening value;
3. Identify the variables that contribute most strongly to predictions so that the resulting model remains interpretable and clinically reviewable; and
4. Produce a reproducible proof-of-concept that can justify, but not substitute for, subsequent external clinical validation.

For this iteration, business success is defined as evidence that the dataset can support a stable and interpretable risk-stratification model rather than evidence that the model is ready for clinical deployment. The analytical

success criteria are therefore set as: (i) ROC–AUC of at least 0.80 under a validation procedure that keeps training and evaluation data separate; (ii) sensitivity/recall of at least 0.80 for the gallstone class at the selected decision threshold; (iii) performance superior to simple baselines such as majority-class prediction and standard logistic regression where possible; and (iv) reporting of calibration, confusion-matrix outcomes, and important predictors in addition to overall accuracy. These thresholds are project benchmarks chosen to make the data-mining goal measurable; they are not clinical guideline thresholds.

2.3 Situation assessment

2.3.1 Available resources and data

The UCI Gallstone dataset originates from a prospective descriptive study conducted at the Internal Medicine Outpatient Clinic of Ankara VM Medical Park Hospital between June 2022 and June 2023 I. Esen et al., 2025; İ. Esen et al., 2024. The source study initially considered 454 participants and excluded individuals who had previously undergone gallbladder surgery, leaving 319 participants for analysis: 161 with gallstones and 158 controls İ. Esen et al., 2024. The near-balanced outcome distribution is advantageous for a binary classification exercise because the model is not forced to learn from a severely under-represented positive class.

The dataset contains demographic variables, comorbidities, anthropometric measurements, bioimpedance measurements, hepatic-fat information, and routine laboratory results. Examples include age, sex, body mass index (BMI), diabetes, hyperlipidaemia, total body water, extracellular and intracellular water, body-fat and lean-mass measures, visceral fat measures, glucose, lipid markers, liver enzymes, creatinine, glomerular filtration rate, C-reactive protein (CRP), haemoglobin, and vitamin D I. Esen et al., 2025; İ. Esen et al., 2024. The UCI repository describes the released data as having no missing values I. Esen et al., 2025; however, this must still be confirmed programmatically before modelling rather than assumed from documentation alone.

The dataset is computationally small and can be analysed on ordinary hardware. All development in this iteration will use Python, with reproducible scripts and notebooks used to record data validation, preprocessing, feature engineering, model fitting, and evaluation. The principal libraries are pandas and NumPy for data handling, scikit-learn for preprocessing, validation, and baseline models, and XGBoost and LightGBM for boosted-tree experiments.

2.3.2 Requirements and assumptions

Several requirements follow from the medical context. First, all predictor transformations must be fitted using training data only so that information from the evaluation set cannot leak into model development. Second, model performance must be reported with class-sensitive metrics—particularly sensitivity, specificity, precision, F1 score, ROC–AUC, and the confusion matrix—rather than accuracy alone. Third, model interpretation is required because a high-performing but opaque model would be difficult to justify as clinical decision support. Finally, the analysis should distinguish prediction from causal inference: an important feature may improve classification without being a causal determinant of gallstone formation.

A critical data-validation requirement is to verify the target encoding directly from the supplied dataset before analysis. The published study describes one numerical coding for gallstone presence, whereas the UCI repository documentation presents the class labels differently I. Esen et al., 2025; İ. Esen et al., 2024. Similarly, repository metadata and the study description should be reconciled for the final number of observations. These discrepancies do not prevent modelling, but failure to resolve them could invert the target labels or create incorrect performance interpretation.

2.3.3 Constraints, risks, and contingencies

The principal constraint is sample size. With only approximately 319 participants and a relatively large number of candidate predictors, flexible models can overfit the training data. This risk will be reduced through stratified resampling, cross-validation, regularisation or restricted model complexity, and comparison with simpler baselines. Hyperparameter tuning and feature selection must occur inside the training/resampling process rather than on the full dataset.

A second constraint is external validity. The records come from a single hospital population in Ankara, and the observed relationships may not transfer directly to other countries, ethnic groups, age structures, or healthcare settings. Consequently, even strong internal validation results should be described as proof-of-concept evidence. External validation on an independent cohort would be required before clinical use.

A third risk concerns measurement availability. Some variables, particularly detailed bioimpedance measurements, may not be collected in every routine clinical setting. To address this, the project should compare a full-feature model with a reduced model based on more routinely available demographic and laboratory variables. If predictive performance remains similar, the reduced model may provide greater practical value.

Other risks include correlated body-composition variables, unstable feature importance, hidden coding errors, and optimistic results caused by repeated experimentation on the same test data. The contingency plan is to preserve a clear evaluation protocol, report uncertainty across resamples, investigate highly correlated variables, and avoid claiming that a small improvement in a single metric represents clinically meaningful superiority.

2.4 Data-mining goals

The data-mining task is a supervised binary-classification problem in which gallstone status is the target variable. The immediate technical goals are to:

1. Audit variable types, ranges, coding, duplicates, and missing values;
2. Explore relationships between gallstone status and demographic, metabolic, laboratory, and body-composition predictors;
3. Establish interpretable baselines using logistic regression and a simple tree-based model;
4. Compare a limited set of nonlinear classifiers such as random forest and gradient boosting using consistent resampling;
5. Tune decision thresholds with particular attention to sensitivity and false-negative errors;
6. Assess probability calibration in addition to discrimination;
7. Evaluate model behaviour across clinically relevant subgroups, especially sex and age groups where sample size permits; and
8. Use interpretable feature-attribution or model-specific importance methods to identify variables that consistently contribute to predictions.

These goals align directly with the business objective: the model is useful only if it can identify higher-risk individuals with acceptable sensitivity, provide a meaningful probability estimate, and explain the factors contributing to that estimate. The strongest model will therefore not automatically be the model with the highest raw accuracy; model selection will consider discrimination, sensitivity, calibration, stability, and interpretability together.

2.5 Project plan for the current iteration

Table 1 summarises the iteration running from 21 September to 16 October 2026. Development is organised around a common Python evaluation framework so that the baseline, data-centric, and model-centric strategies can be compared fairly using the same data partitions and metrics.

Table 1: Iteration plan for Python development and evaluation, 21 September–16 October 2026.

Dates	**Main activities**	**Expected output**
21–23 September	Confirm the business objective and success criteria; audit target coding, variable types, missing values, duplicates, ranges, and provenance in Python.	Validated problem statement, data dictionary, and data-quality log.
24–26 September	Complete exploratory analysis and implement a leakage-safe Python pipeline with fixed stratified 70/30 partitions and repeated seeds; establish logistic-regression and tree baselines.	Reproducible baseline workflow, saved split membership, and benchmark metrics.
27 September–3 October	Apply the *data-centric strategy*: correct data-quality issues, assess transformations and outliers, engineer clinically meaningful features, compare full and reduced feature sets, and perform feature selection within cross-validation.	Audited feature sets and evidence of whether data improvements increase validation performance and stability.
27 September–6 October	In parallel, apply the *model-centric strategy*: compare logistic regression, random forest, XGBoost, LightGBM, and stacking; tune promising models without accessing the test set.	Tuned candidate models and a cross-validated comparison table.
7–10 October	Evaluate alternative validation strategies using five-seed stratified holdout with training-only three-fold cross-validation; assess threshold choice, calibration, sensitivity, specificity, ROC–AUC, F1, and log loss.	Robustness analysis showing the effect of evaluation strategy and decision threshold.
11–13 October	Conduct error analysis, subgroup checks, feature-importance analysis, and comparison of data-centric, model-centric, and combined strategies.	Error profiles, interpretation outputs, and a defensible preferred approach.
14–16 October	Re-run the selected workflow, verify reproducibility, consolidate results and limitations, and finalise the report and recommendations.	Reproducible Python artefacts and final iteration report.

2.5.1 Development and data-mining plan

The current iteration is scheduled from 21 September to 16 October 2026. All development will be completed in Python. The work begins with a reproducible baseline and then separates improvement activities into two complementary strategies. The data-centric strategy improves the quality and representation of the input data through validation, transformation, feature engineering, feature selection, and reduced-feature experiments. The model-centric strategy holds the prepared data and evaluation rules constant while comparing model families, hyperparameters, ensembles, calibration, and classification thresholds. A final combined strategy uses only improvements supported by validation evidence.

Figure 2 presents the project timeline. The same saved data partitions and metrics will be used across strategies to ensure that any observed improvement is attributable to the tested change rather than to a different evaluation setup.

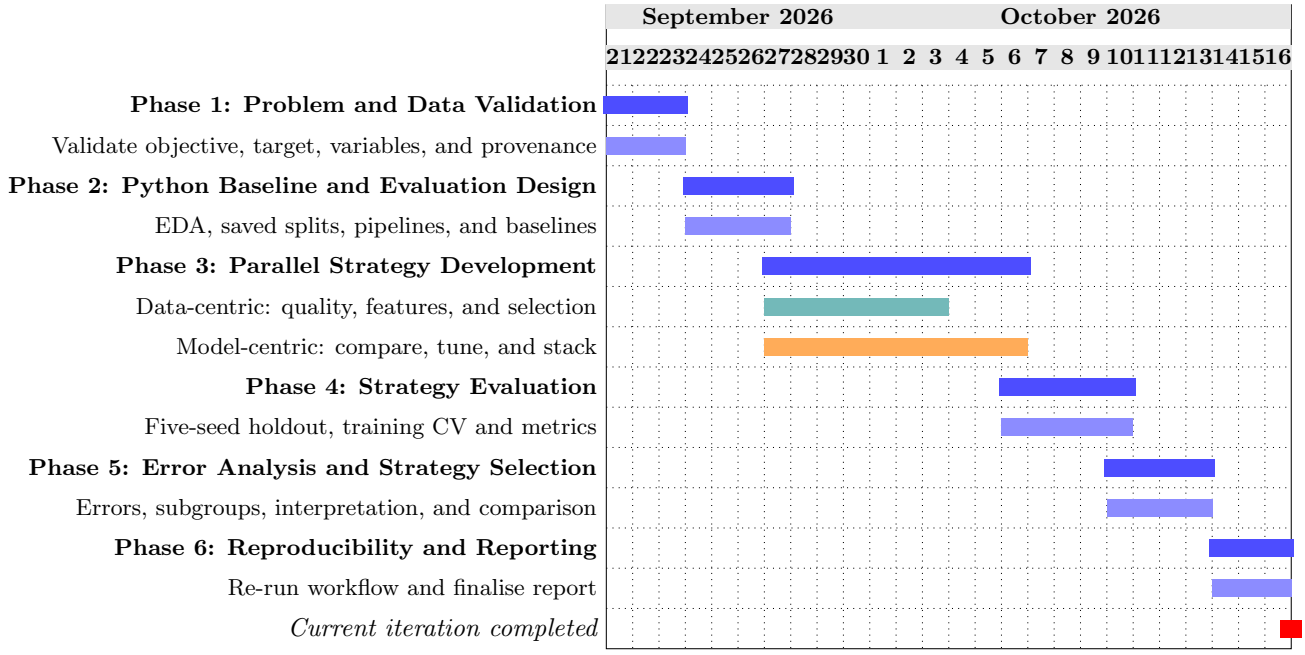


Figure 2: Python development and evaluation plan for the current gallstone prediction iteration, 21 September–16 October 2026.

The baseline strategy provides the reference result against which subsequent changes are assessed. In the data-centric stage, models are deliberately kept stable while changes to data quality, transformations, engineered variables, and feature subsets are evaluated. In the model-centric stage, the selected data representation is held stable while algorithms and hyperparameters are varied. Preprocessing and feature selection remain inside each resampling fold to prevent leakage. Repeated holdout evaluation measures variation across partitions. Three-fold cross-validation within each training partition supplies tuning and algorithm-selection evidence. The preferred approach will be selected using sensitivity, specificity, ROC–AUC, F1, calibration, stability, interpretability, and implementation complexity rather than accuracy alone.

Figure 3 clarifies the experimental pipeline. The untouched test partition is used only after the data-centric and model-centric choices have been made from training-only cross-validation evidence.

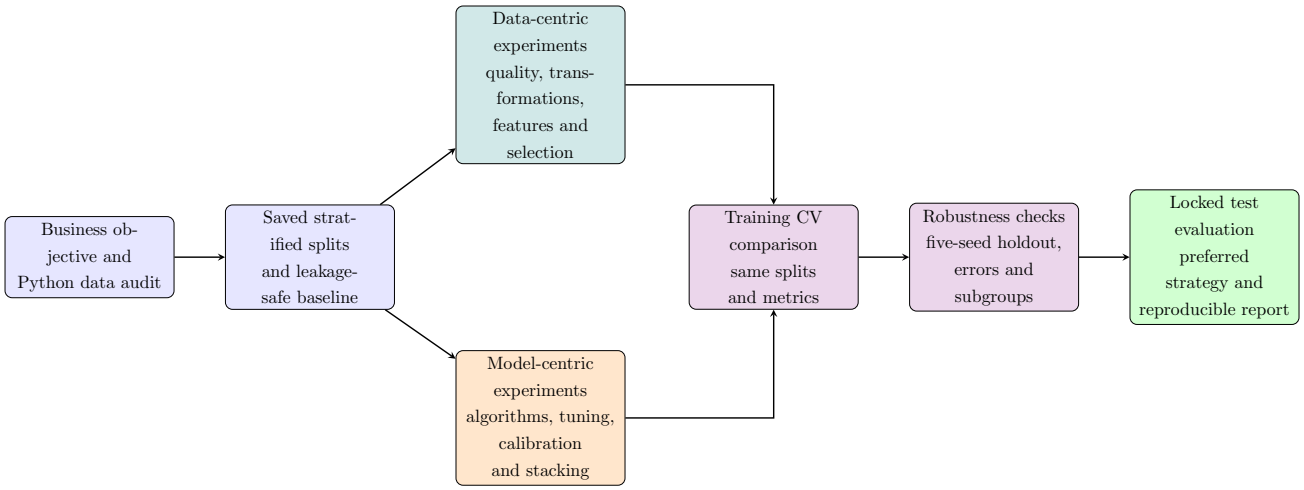


Figure 3: Python project pipeline for comparing data-centric, model-centric, and combined gallstone-prediction strategies.

2.5.2 Report writing plan

Report writing will be carried out progressively throughout the current iteration rather than being postponed until the modelling work is complete. Early sections describing the problem, dataset, and methodology will be drafted while the corresponding analytical tasks are being completed. Results, discussion, limitations, and recommendations will then be added as model outputs become available. This approach allows the report to be updated continuously and reduces the risk of documentation being disconnected from the implemented analysis.

Figure 4 presents the report-writing schedule for 21 September–16 October 2026.



Figure 4: Day-to-day report-writing plan for the gallstone prediction project, 21 September–16 October 2026.

The completion criterion for this iteration is therefore a transparent comparison of feasible predictive models and a defensible conclusion about whether this dataset supports early gallstone-risk stratification. Clinical deployment, treatment decisions, and replacement of diagnostic imaging are explicitly outside the scope of the present iteration.

3 Data understanding

3.1 Initial Data Collection

The dataset used in this project was obtained from the UCI Machine Learning Repository and originates from the clinical study conducted by İ. Esen et al. (2024). The original data were collected prospectively from patients attending the Internal Medicine Outpatient Clinic of Ankara VM Medical Park Hospital, Turkey, between June 2022 and June 2023 (I. Esen et al., 2025; İ. Esen et al., 2024).

The original study initially collected information from 454 participants. Records from individuals who had previously undergone gallbladder surgery were excluded, resulting in a final published dataset of 319 participants. Among these, 161 participants were diagnosed with gallstone disease and 158 formed the control group (İ. Esen et al., 2024). Data collection included demographic and clinical characteristics, comorbidities, laboratory test results, and body-composition measurements obtained through bioimpedance analysis. The resulting dataset contains 38 predictive attributes relating to factors such as age, BMI, diabetes, lipid profiles, liver-function markers, glucose levels, body water, muscle mass, and body-fat distribution/

No additional primary data were collected for the current project; therefore, this analysis represents secondary use of the published clinical dataset. The UCI repository reports no missing values. However, an issue identified during initial data collection and validation is an inconsistency in the published metadata: the repository description reports 319 participants, while one metadata field lists 320 instances. Consequently, the downloaded dataset's row count and target coding will be independently verified before further analysis.

The dataset was downloaded in tabular form and will be retained in its original form as a raw-data copy before any cleaning or transformation is performed.

3.2 Data Description

The dataset contains one binary target variable, *Gallstone Status*, and 38 predictor variables covering demographic characteristics, comorbidities, anthropometric measurements, body-composition indicators, and laboratory measurements. The predictors include both categorical and numerical variables, allowing the project to investigate multiple clinical and physiological factors that may be associated with gallstone disease.

According to the dataset documentation, no missing values are reported. However, the variable types, class coding, ranges, and missing-value patterns will be independently verified during the data-quality assessment before modelling. In particular, categorical variables stored numerically will be converted to appropriate factor or categorical representations to prevent their codes from being interpreted as continuous quantities.

Table 2 summarises the variables contained in the dataset.

Table 2: Data dictionary for the Gallstone dataset.

Variable	Role	Type	Demo.	Description	Units	Missing
Gallstone Status	Target	Binary	–	Presence or absence of gallstones	–	No
Age	Feature	Integer	Age	Age of the participant	years	No
Gender	Feature	Categorical	Gender	Gender of the participant	–	No
Comorbidity	Feature	Categorical	–	Number of concomitant medical conditions	–	No
Coronary Artery Disease (CAD)	Feature	Binary	–	Presence of coronary artery disease	–	No
Hypothyroidism	Feature	Binary	–	Presence of an underactive thyroid gland	–	No

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Variable	Role	Type	Demo.	Description	Units	Missing
Hyperlipidemia	Feature	Binary	–	Presence of elevated blood lipid levels	–	No
Diabetes Mellitus (DM)	Feature	Binary	–	Presence of diabetes mellitus	–	No
Height	Feature	Integer	–	Height of the participant	–	No
Weight	Feature	Continuous	–	Body weight of the participant	–	No
Body Mass Index (BMI)	Feature	Continuous	–	Weight relative to height	–	No
Total Body Water (TBW)	Feature	Continuous	–	Total amount of water in the body	–	No
Extracellular Water (ECW)	Feature	Continuous	–	Amount of water outside body cells	–	No
Intracellular Water (ICW)	Feature	Continuous	–	Amount of water contained within body cells	–	No
Extracellular Fluid/Total Body Water (ECF/TBW)	Feature	Continuous	–	Ratio of extracellular fluid to total body water	–	No
Total Body Fat Ratio (TBFR)	Feature	Continuous	–	Proportion of total body mass consisting of fat	%	No
Lean Mass (LM)	Feature	Continuous	–	Proportion of lean body mass	%	No
Body Protein Content (Protein)	Feature	Continuous	–	Estimated proportion of body protein	%	No
Visceral Fat Rating (VFR)	Feature	Integer	–	Rating of visceral fat surrounding internal organs	–	No
Bone Mass (BM)	Feature	Continuous	–	Estimated bone mass	–	No
Muscle Mass (MM)	Feature	Continuous	–	Estimated amount of muscle mass	–	No
Obesity	Feature	Continuous	–	Estimated degree of excess adiposity	%	No
Total Fat Content (TFC)	Feature	Continuous	–	Estimated total amount of body fat	–	No
Visceral Fat Area (VFA)	Feature	Continuous	–	Estimated area of visceral adipose tissue	–	No
Visceral Muscle Area (VMA)	Feature	Continuous	–	Estimated visceral muscle measurement	kg	No
Hepatic Fat Accumulation (HFA)	Feature	Categorical	–	Degree of fat accumulation in the liver	–	No
Glucose	Feature	Continuous	–	Blood glucose level	–	No
Total Cholesterol (TC)	Feature	Continuous	–	Total blood cholesterol level	–	No
Low Density Lipoprotein (LDL)	Feature	Continuous	–	Low-density lipoprotein cholesterol level	–	No
High Density Lipoprotein (HDL)	Feature	Continuous	–	High-density lipoprotein cholesterol level	–	No
Triglyceride	Feature	Continuous	–	Triglyceride concentration in the blood	–	No
Aspartate Aminotransferase (AST)	Feature	Continuous	–	Liver enzyme measurement	–	No
Alanine Aminotransferase (ALT)	Feature	Continuous	–	Liver enzyme measurement	–	No

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Table 2 – continued from previous page

Variable	Role	Type	Demo.	Description	Units	Missing
Alkaline Phosphatase (ALP)	Feature	Continuous	–	Enzyme associated primarily with liver and bone	–	No
Creatinine	Feature	Continuous	–	Indicator associated with kidney function	–	No
Glomerular Filtration Rate (GFR)	Feature	Continuous	–	Estimated kidney filtration rate	–	No
C-Reactive Protein (CRP)	Feature	Continuous	–	Marker associated with systemic inflammation	–	No
Hemoglobin (HGB)	Feature	Continuous	–	Concentration of oxygen-carrying haemoglobin	–	No
Vitamin D	Feature	Continuous	–	Blood vitamin D level	–	No

3.2.1 Categorical Variable Coding

Several categorical variables are represented by integer codes in the raw dataset. These values represent categories rather than continuous numerical quantities and therefore require appropriate treatment during preprocessing. The documented class labels are:

- **Gallstone Status:** 0 = gallstones present; 1 = gallstones absent.
- **Gender:** 0 = male; 1 = female.
- **Comorbidity:** 0 = no comorbidities; 1 = one comorbid condition; 2 = two comorbid conditions; 3 = three or more comorbid conditions.
- **Coronary Artery Disease:** 0 = no; 1 = yes.
- **Hypothyroidism:** 0 = no; 1 = yes.
- **Hyperlipidemia:** 0 = no; 1 = yes.
- **Diabetes Mellitus:** 0 = no; 1 = yes.
- **Hepatic Fat Accumulation (HFA):** 0 = no fat accumulation; 1 = Grade 1 (mild); 2 = Grade 2 (moderate); 3 = Grade 3 (severe); 4 = Grade 4 (very severe).

Although these categories are encoded numerically, they do not all have the same statistical interpretation. Binary variables such as diabetes and coronary artery disease can be represented as indicator variables, whereas *Comorbidity* and *Hepatic Fat Accumulation* are ordinal variables because their categories represent increasing levels of severity or burden. Their treatment during modelling will therefore be determined explicitly during the data-preparation stage.

3.3 Data exploration

Exploratory data analysis (EDA) was conducted on the raw dataset before any transformation or model training. The purpose was to understand the distribution of the target variable, examine the characteristics of the predictors, identify potential outliers or skewed variables, and investigate relationships that may influence subsequent data preparation and modelling.

3.3.1 Target Variable Distribution

Figure 5 shows the distribution of the target variable, *Gallstone Status*. Based on the documented class encoding, 0 represents the presence of gallstones and 1 represents the absence of gallstones.

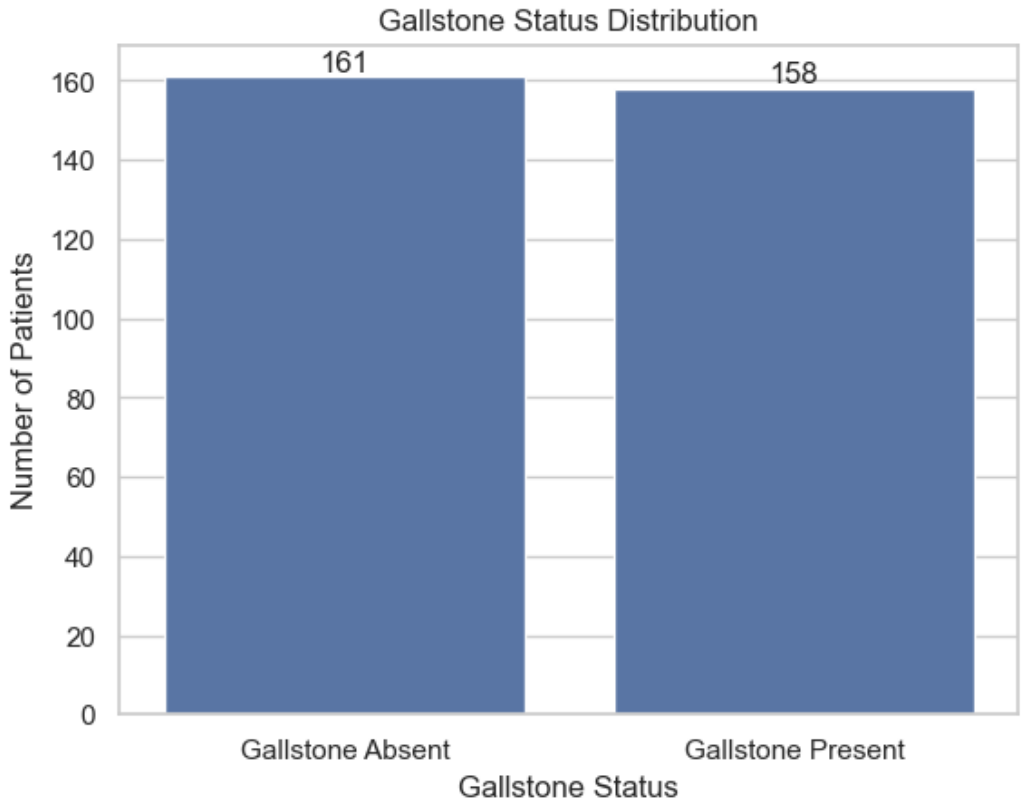


Figure 5: Distribution of Gallstone Status in the raw dataset.

The dataset contains 161 participants with gallstones and 158 participants without gallstones. The two classes therefore account for approximately 50.5% and 49.5% of the observations, respectively. This indicates that the classification problem is almost perfectly balanced. This is an important characteristic for the subsequent modelling stage. Severe class imbalance is unlikely to be a major source of bias, and resampling techniques such as oversampling or undersampling are not immediately required. Nevertheless, stratified data partitioning will be used when dividing the observations into training and testing sets to preserve the approximately equal class proportions. Model evaluation will also consider sensitivity, specificity, F1-score and ROC-AUC rather than accuracy alone.

3.3.2 Distribution of Continuous Features

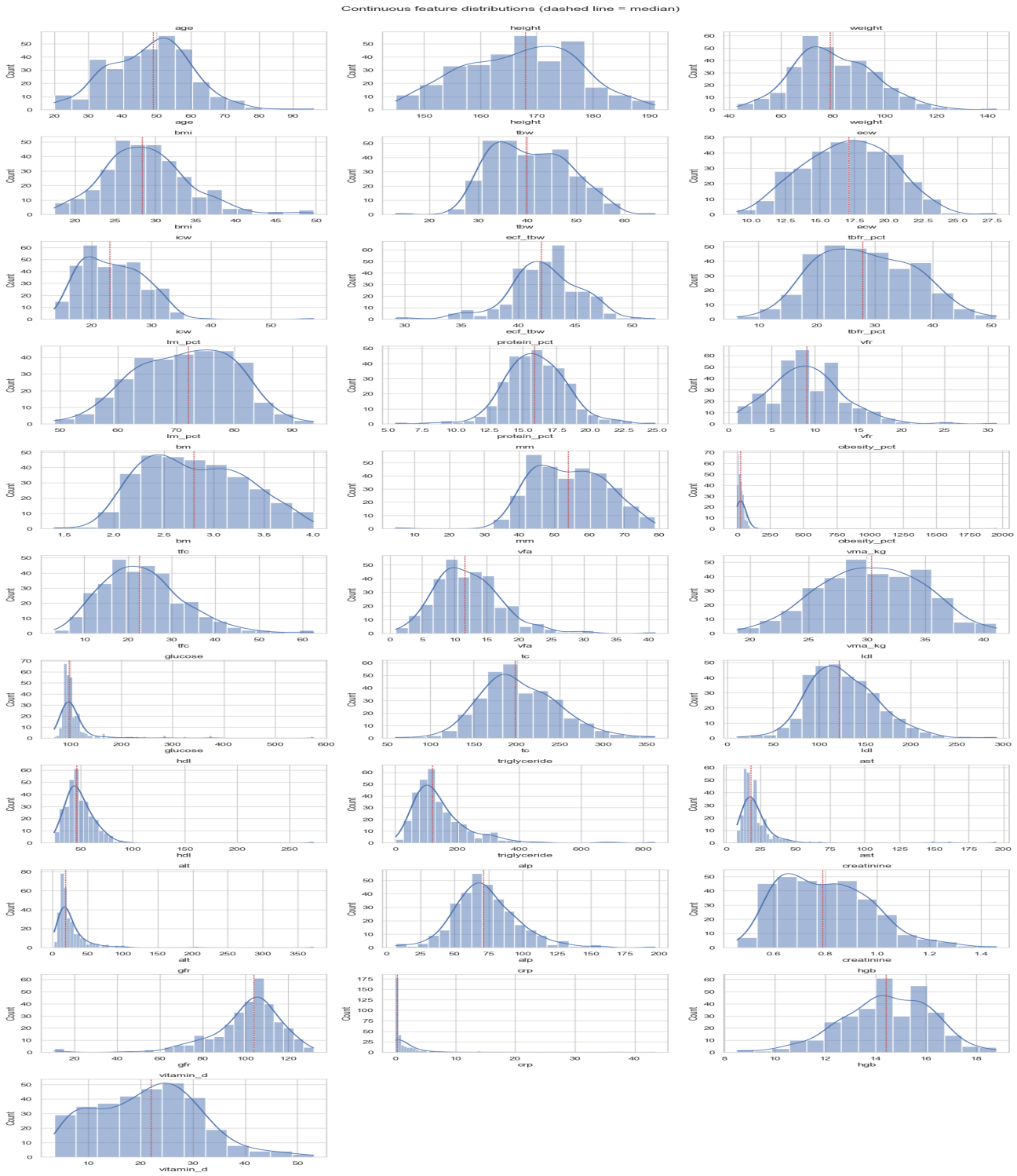


Figure 6: Distribution of continuous variables in the raw Gallstone dataset. The dashed vertical line represents the median.

Figure 6 presents the distributions of the continuous variables before any transformation or outlier treatment. The plots show considerable variation in distributional shape across the clinical and body-composition measurements.

Several variables, including age, height, BMI, extracellular water (ECW), lean mass, creatinine, haemoglobin, and visceral muscle area, display relatively smooth and approximately unimodal distributions. In contrast, variables such as glucose, triglyceride, AST, ALT, CRP, visceral fat rating (VFR), visceral fat area (VFA), and total fat content (TFC) show noticeable right-skewness, with most observations concentrated at lower values and a small

number of substantially larger measurements.

The distributions also reveal several extreme observations, particularly in metabolic and laboratory variables. These values require further assessment during the data-quality stage to determine whether they represent valid clinical measurements or potential data errors. They will not be removed automatically, as unusual values may contain clinically meaningful information relevant to gallstone risk.

Overall, this exploration indicates that the predictors differ substantially in both scale and distribution. Consequently, standardisation and possible transformations of strongly skewed variables will be considered during data preparation, depending on the modelling technique used. The next stage of the analysis compares these predictors with *Gallstone Status* to identify variables that may provide useful discriminatory information for the classification models.

3.3.3 Distribution of categorical features and Association with Gallstone Status

The categorical predictors were examined by comparing both the class counts and the proportion of participants with gallstones within each category. This provides an initial indication of whether particular categorical variables may contain useful discriminatory information for the prediction task.

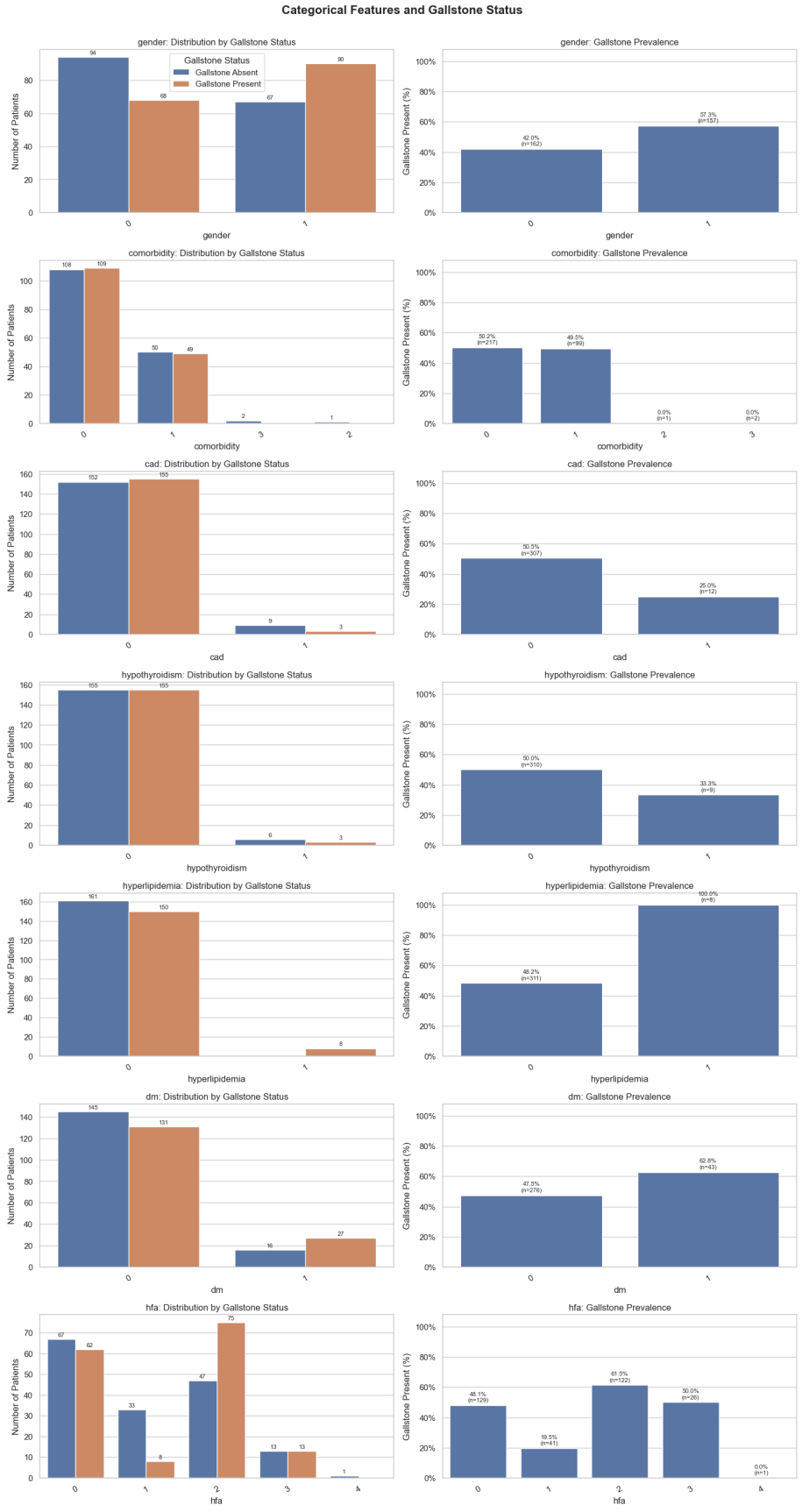


Figure 7: Distribution of categorical predictors by Gallstone Status and the corresponding gallstone prevalence within each category.

Figure 7 suggests that the predictive signal varies considerably across the categorical variables. Gender shows a moderate difference between categories: the gallstone prevalence is approximately 42.0% for category 0 and

57.3% for category 1, indicating that gender may provide some discriminatory information.

Diabetes mellitus (`dm`) also shows a potentially useful difference. Gallstone prevalence is approximately 47.5% among participants in category 0 compared with 62.8% in category 1. A similar pattern is visible for `hfa`, where prevalence varies across categories, most notably between category 1 (19.5%) and category 2 (61.5%). This suggests that these variables may contribute information useful for separating the two outcome classes.

In contrast, `comorbidity` shows little separation in the two major categories, with gallstone prevalence close to 50% in both. This indicates limited univariate discriminatory value for this variable. `cad` and `hypothyroidism` also show lower prevalence among category 1 than category 0, although the number of observations in their positive categories is small.

The strongest apparent difference is observed for `hyperlipidemia`, where all eight participants in category 1 have gallstones. However, this result should be interpreted cautiously because the category contains very few observations. Similar caution is required for rare levels of `comorbidity` and `hfa`, where prevalence estimates are based on only one or two participants and are therefore highly unstable.

Overall, the categorical EDA provides evidence of some class-related signal, particularly for gender, diabetes mellitus, and selected HFA categories. However, no single categorical predictor provides a sufficiently strong or stable separation between participants with and without gallstones. The results therefore support retaining these variables for multivariable modelling, where their combined effects and interactions with continuous clinical and body-composition measurements can be evaluated.

3.3.4 Correlation Analysis

Figure 8 presents the Spearman correlationmatrix for the dataset. Spearman correlation was selected because several continuous variables are skewed and contain extreme observations.

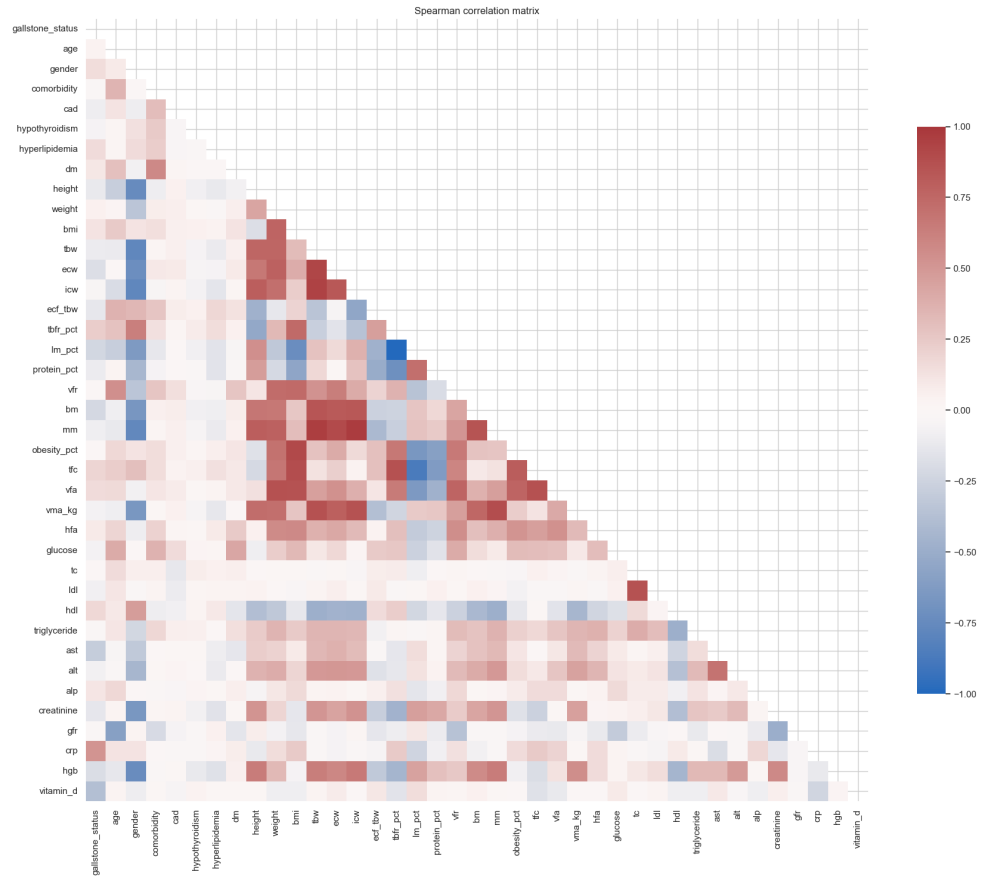


Figure 8: Spearman correlation matrix for the Gallstone dataset visualization.

The matrix shows several groups of strongly related predictors. In particular, body-composition variables such as weight, BMI, TBW, ECW, ICW, muscle mass, body-fat measures, and visceral-fat measures show substantial correlation. Laboratory variables also contain related groups, including TC-LDL, AST-ALT, and the inverse relationship between creatinine and GFR. These patterns indicate potential redundancy and multicollinearity among some predictors.

In contrast, individual correlations with *Gallstone Status* are generally weaker. This suggests that gallstone status is unlikely to be explained by a single predictor and supports the use of multivariable classification models that combine demographic, body-composition, and laboratory information.

These findings will guide the data-preparation stage. Highly correlated variables will be reviewed during feature selection, particularly for logistic regression, while tree-based models will also be considered because they are less sensitive to multicollinearity.

3.3.5 Structure Among Body-Composition Variables

Some body-composition variables, including total body water (TBW), bone mass (BM), and muscle mass (MM), appear to have broader or mildly multi-peaked distributions. This may indicate underlying differences between participant subgroups, for example differences in sex, body size, or body composition. At this stage, however, the univariate plots alone are insufficient to determine the cause. Furthermore, several predictors measure related physiological concepts. For example, BMI, TBFR, TFC, VFR and VFA all contain information relating to body fat or adiposity, while TBW, ECW and ICW describe different components of body watxer. Consequently, correlation analysis will be conducted in the next stage to determine whether some predictors contain highly overlapping information. Highly correlated variables will be reviewed before modelling to reduce unnecessary redundancy and potential multicollinearity, particularly for regression-based classifiers.

Overall, the EDA provides the foundation for the next data-preparation phase by identifying class balance, distributional characteristics, outliers, potentially informative predictors, and redundant variables. These findings will subsequently guide preprocessing, feature selection, model selection, and evaluation.

3.4 Data Quality

Data quality was assessed before any transformation or model development to ensure that errors in the raw data would not be carried into the predictive models. The verification focused on missing values, duplicate records, categorical coding, numerical ranges, and potential outliers.

3.4.1 Missing Values and Completeness

The dataset documentation reports no missing values, and this was confirmed during the initial inspection of the dataset. Therefore, no imputation was required at this stage. Nevertheless, missing-value checks will remain part of the preprocessing workflow to ensure that any values introduced during transformation or partitioning are detected before modelling.

3.4.2 Potential Outliers and Data Quality Issues

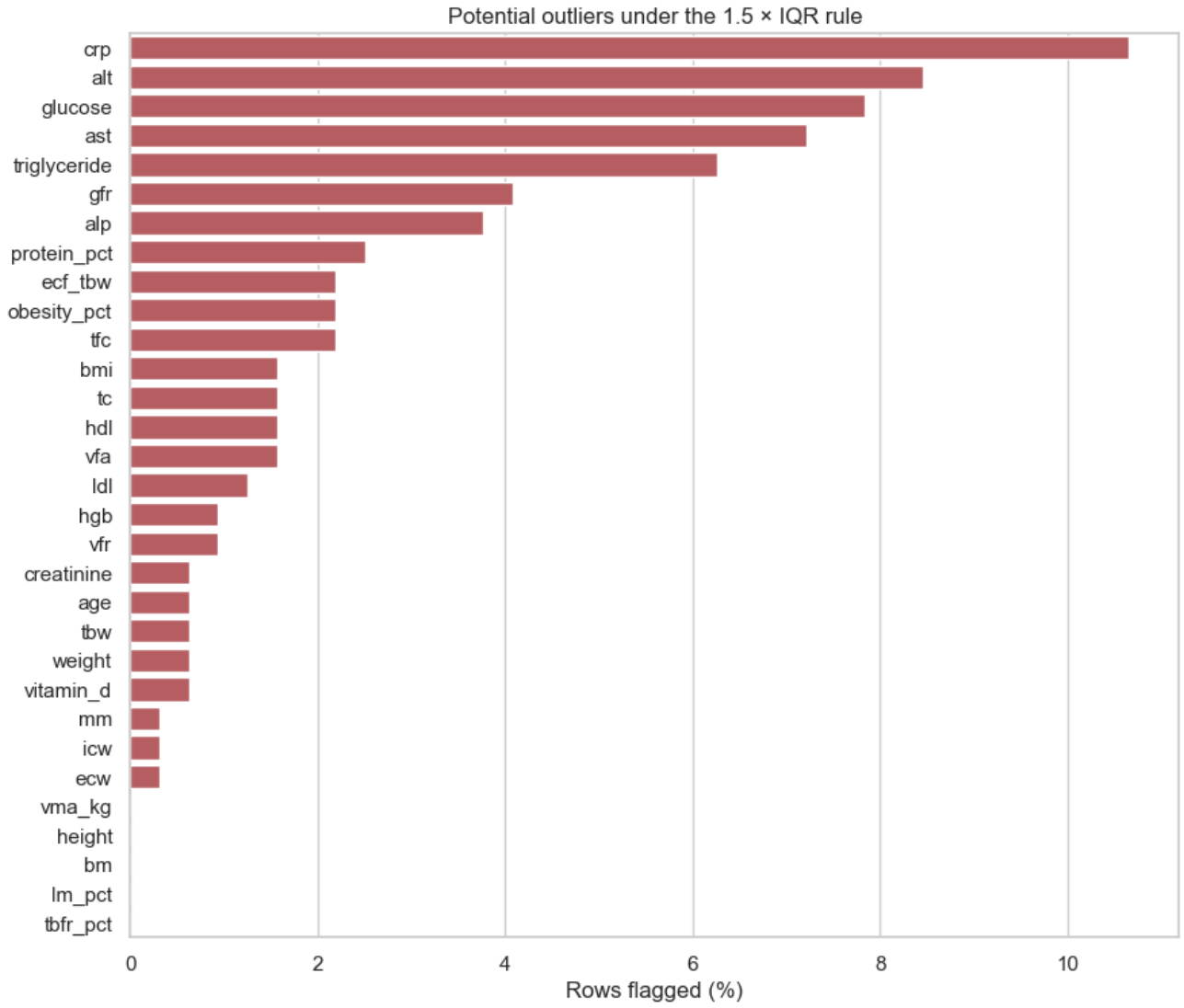


Figure 9: Outlier analysis using the 1.5×IQR method.

Figure 9 identifies several observations that require further investigation. Most notably, the *Obesity* variable contains an extreme value of approximately 2000, which is substantially larger than the remaining observations. Given the magnitude of this difference, the value should be verified against the original dataset and variable definition to determine whether it represents a valid measurement, a unit inconsistency, or a data-entry error.

Potential upper-tail outliers are also observed in variables including glucose, HDL, triglyceride, AST, ALT, CRP, and intracellular water (ICW). However, observations identified by the 1.5×IQR rule should not automatically be treated as erroneous, particularly in clinical data where extreme measurements may represent genuine patient conditions. Therefore, these observations will be reviewed during data preparation before any correction, transformation, or removal is applied.

Valid extreme observations will be retained where appropriate to preserve clinically meaningful variation in the dataset. Only values supported by evidence of coding, measurement, or data-entry errors will be considered for correction or exclusion. This approach reduces the risk of removing informative clinical cases and introducing bias into subsequent predictive models.

3.4.3 Dimensionality Reduction Visualisation and Data Understanding

Dimensionality-reduction techniques were used to obtain a more compact view of the relationships among the continuous measurements and to assess whether participants with and without gallstones exhibit distinct multivariate patterns. Two complementary approaches were considered: Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE). PCA provides a linear representation of the dominant sources of variance in the data, whereas t-SNE is a nonlinear technique designed to preserve local neighbourhood relationships in a low-dimensional representation (van der Maaten & Hinton, 2008).

Principal Component Analysis PCA was first applied to the continuous measurements to examine the main directions of variation in the dataset. The first principal component (PC1) explained 27.7% of the total variance, while the second principal component (PC2) explained a further 20.0%. Together, the first two components therefore accounted for approximately 47.7% of the variance in the continuous feature space.

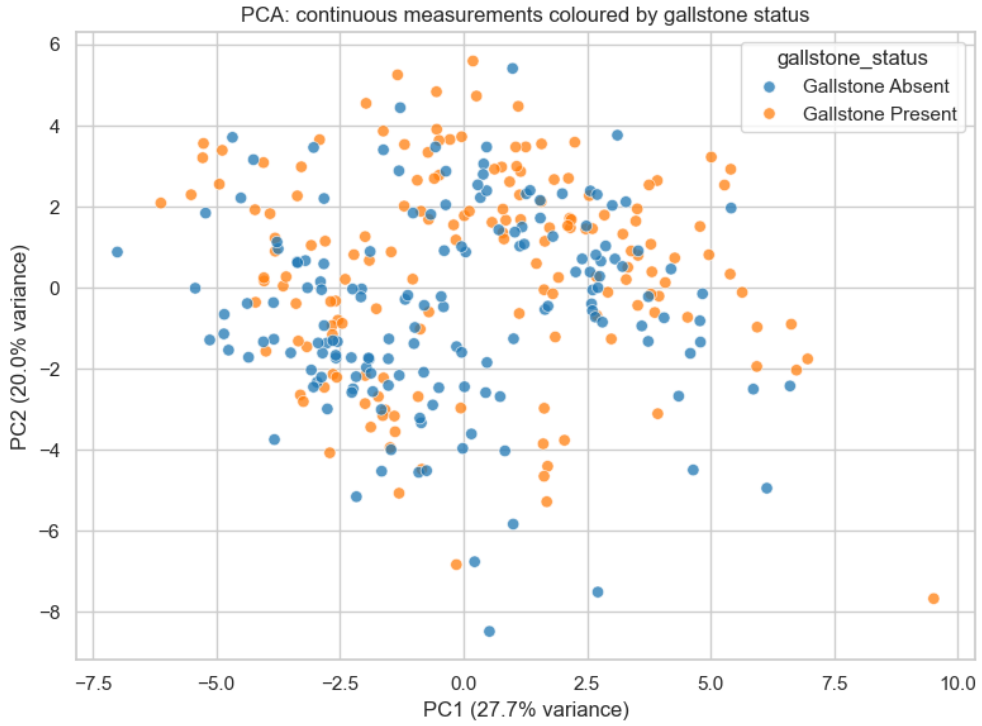


Figure 10: Two-dimensional PCA projection of the continuous measurements, coloured by Gallstone Status. PC1 and PC2 explain 27.7% and 20.0% of the total variance, respectively.

As shown in Figure 10, observations from the *Gallstone Present* and *Gallstone Absent* groups overlap substantially in the first two principal components. No distinct linear separation between the two outcome groups is apparent. This indicates that the largest sources of variation in the continuous measurements are not themselves equivalent to the variation that separates gallstone status.

Nevertheless, the projection reveals meaningful structure within the measurements. Participants are distributed across several regions of the PCA space, suggesting heterogeneity in body-composition, anthropometric, and related clinical characteristics. A small number of observations also lie relatively far from the central concentration of the data. These observations are consistent with the unusual or extreme measurements identified during the earlier univariate exploration and were retained for further investigation rather than being removed solely on the basis of their position in the PCA projection.

To understand which variables contribute most strongly to the two principal components, the corresponding PCA loadings were also examined.

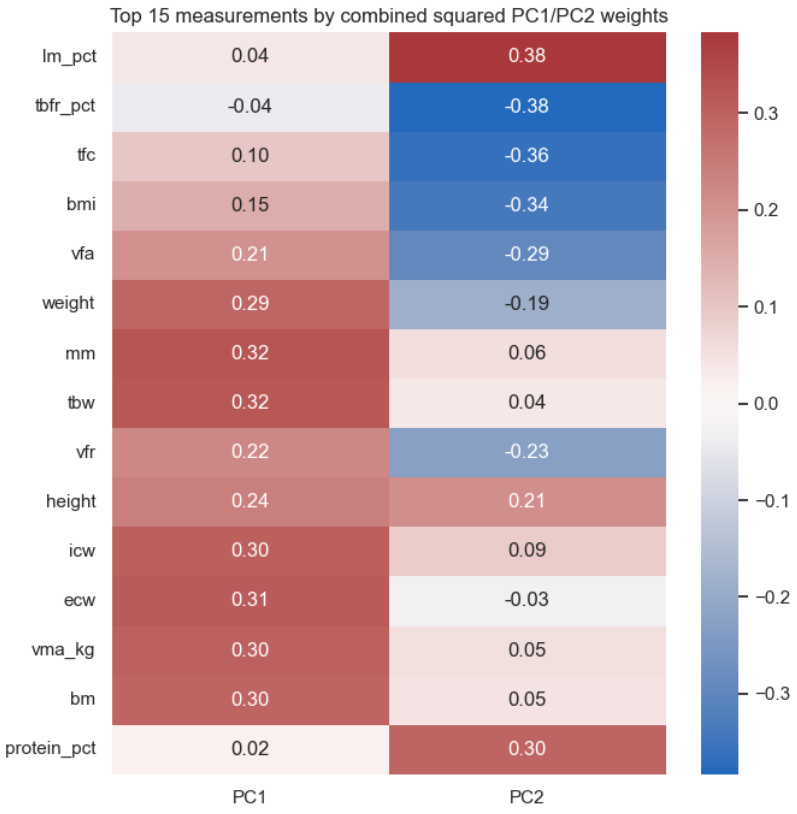


Figure 11: Loadings of the 15 continuous measurements with the largest combined squared contributions to PC1 and PC2. Positive and negative values indicate the direction of association with each principal component.

Figure 11 shows that PC1 is influenced primarily by several body-size and body-composition measurements. Muscle mass (`mm`), total body water (`tbw`), extracellular water (`ecw`), intracellular water (`icw`), visceral muscle area (`vma_kg`), bone mass (`bm`), weight, and height all have positive loadings on PC1. This suggests that PC1 broadly represents variation associated with overall body size and lean/body-composition characteristics.

PC2 captures a different contrast. Lean-mass percentage (`lm_pct`) and protein percentage (`protein_pct`) have strong positive loadings, whereas total body-fat percentage (`tbfr_pct`), total fat content (`tfc`), BMI, visceral fat area (`vfa`), and visceral fat rating (`vfr`) have negative loadings. PC2 can therefore be interpreted as reflecting a contrast between lean-body composition and fat-related measurements. These loading patterns provide additional insight into the correlations among the original measurements and help explain the structure observed in the PCA projection.

t-SNE Visualisation Because PCA represents only linear combinations of the original variables, a nonlinear t-SNE analysis was additionally performed to examine whether local neighbourhood structures could reveal patterns that were not visible in the PCA projection. Multiple perplexity values were evaluated rather than relying on a single t-SNE configuration.

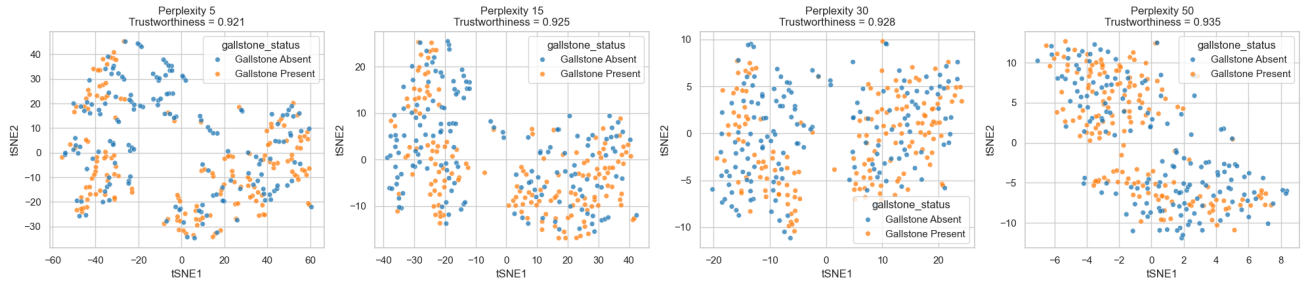


Figure 12: Two-dimensional t-SNE projections of the continuous measurements using perplexity values of 5, 15, 30, and 50. Points are coloured according to Gallstone Status. Trustworthiness scores are reported for each configuration.

Figure 12 shows that the overall appearance of the t-SNE representation changes as the perplexity parameter increases. Lower perplexity values, particularly 5, emphasise very local neighbourhoods and produce a more fragmented representation. At larger perplexities, the observations form broader and more coherent structures.

The trustworthiness score increased from 0.921 at perplexity 5 to 0.935 at perplexity 50, indicating that the higher-perplexity representations preserved the original local neighbourhood relationships slightly more effectively. However, the purpose of this metric is to assess preservation of neighbourhood structure rather than separation of the target classes.

Across all perplexity settings, participants with and without gallstones remain substantially intermixed. Although some local regions contain a greater concentration of one class than the other, there is no stable class-specific cluster or simple visual boundary that consistently separates *Gallstone Present* from *Gallstone Absent*. The same general conclusion is therefore observed in both the linear PCA representation and the nonlinear t-SNE representations.

These findings suggest that gallstone status is unlikely to be explained by a single dominant measurement or by a simple low-dimensional boundary. Instead, the predictive information may be distributed across several demographic, body-composition, metabolic, and biochemical variables and may involve interactions or nonlinear relationships among them. This observation motivates the subsequent comparison of multivariable classification models.

Importantly, both PCA and t-SNE were used here as exploratory tools for data understanding rather than as measures of predictive performance. Visual overlap between the classes does not imply that the target is unpredictable, since discriminative information may exist in dimensions that are not represented by the first two principal components or by the two-dimensional t-SNE embedding. Model performance is therefore assessed separately using the validation and test partitions.

4 Data Preparation

The purpose of the data-preparation stage was to transform the raw Gallstone dataset into a consistent, traceable, and modelling-ready dataset. Decisions made during this stage were informed by the business objective, findings from the exploratory analysis, data-quality checks, and the technical requirements of the modelling pipeline.

All data-preparation procedures were implemented using an open-source Python workflow. The main libraries used were `pandas` and `NumPy` for data manipulation and validation, `SciPy` for statistical transformations, `scikit-learn` for data partitioning and preprocessing, and `Matplotlib` and `Seaborn` for diagnostic visualisation.

Using Python rather than a graphical modelling environment provides a script-based and reproducible workflow in which each preprocessing operation can be inspected, rerun, and consistently applied to subsequent modelling experiments.

4.1 Data Selection

The modelling dataset contains *Gallstone Status* as the target variable and the available demographic, comorbidity, body-composition, and laboratory measurements as candidate predictors. Variables were initially retained rather than removed solely on the basis of weak univariate relationships because the exploratory analysis suggested that gallstone status may depend on combinations of multiple patient characteristics.

Data selection was performed programmatically using Python. The dataset was loaded into a `pandas DataFrame`, after which column names, data types, unique values, distributions, and summary statistics were inspected. Candidate predictors were retained when they had plausible clinical relevance and sufficient data quality for subsequent modelling.

Variables containing unusual values, strong skewness, or high correlations with other predictors were not immediately removed. Instead, these characteristics were investigated during exploratory analysis and preprocessing. In particular, highly correlated predictors were reviewed during feature selection because multicollinearity may influence coefficient stability in models such as logistic regression.

No patient-identifying information or variables without potential predictive value were required for model development. Identifier fields used for data traceability were retained during data processing but excluded from the feature matrix supplied to the predictive models.

4.2 Data Cleaning

Data cleaning was guided by the issues identified during the earlier data-quality assessment. The Python workflow systematically evaluated missing values, duplicated records, invalid categorical codes, numerical ranges, data types, and potential extreme observations.

Completeness was assessed using functions such as `DataFrame.isna()`, while duplicate observations and field-level consistency were examined using standard `pandas` operations. Summary statistics and value-frequency tables were also generated to identify unexpected values and inconsistencies.

No missing values were identified in the original dataset; therefore, no imputation was required before modelling. Nevertheless, missing-value checks were retained as part of the preprocessing pipeline so that the workflow can identify incomplete observations if the dataset is modified or extended in future iterations.

Categorical variables were checked against their documented coding ranges. Binary and ordinal fields stored numerically were treated according to their clinical meaning rather than automatically interpreted as continuous measurements.

Potential outliers were identified during exploratory analysis using the 1.5×IQR rule:

$$\text{Lower Bound} = Q_1 - 1.5(Q_3 - Q_1),$$

$$\text{Upper Bound} = Q_3 + 1.5(Q_3 - Q_1).$$

Observations outside these limits were reviewed rather than automatically removed. This distinction is important for clinical data because statistically extreme observations may nevertheless correspond to genuine patient measurements.

In particular, the unusually large observation identified in the *Obesity* variable was reviewed as a possible data-quality issue. Similarly, extreme observations in glucose, triglyceride, AST, ALT, CRP, and other laboratory measurements were retained when they represented plausible clinical values. Records would only be corrected or excluded if there were sufficient evidence of a coding, unit, or data-entry error.

Column names were also standardised programmatically to remove unnecessary spaces and special characters and to provide consistent variable names throughout the analysis.

4.2.1 Cleaning and Sanity-Checking Workflow

The data-cleaning process was implemented as a reproducible sequence of Python operations. The main stages were:

1. verify the number of observations and variables;
2. inspect data types and expected variable roles;
3. identify missing and blank values;
4. detect duplicate observations;
5. validate categorical coding ranges;
6. examine descriptive statistics for continuous measurements;
7. identify potential extreme observations;
8. standardise column names and field representations; and
9. confirm the dimensions and integrity of the resulting analytical dataset.

This workflow provided explicit quality-control checkpoints while preserving the original observations whenever no clear data-quality issue was present.

4.2.2 Skewness Assessment and Transformation of Continuous Features

The distributions of the continuous predictors were further examined using sample skewness to identify variables with substantial departures from symmetry. Absolute skewness values greater than or equal to 1 were treated as an indication of strong skewness and were selected for further transformation assessment.

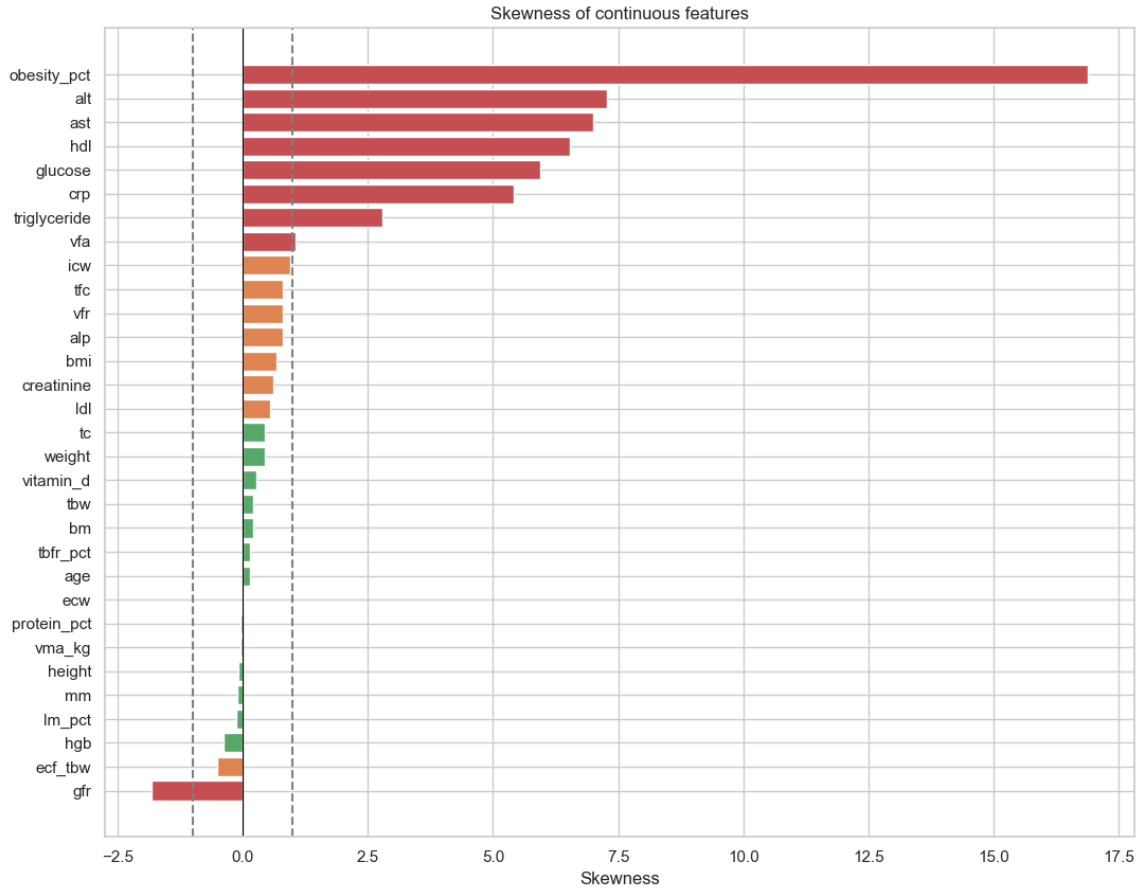


Figure 13: Skewness of the continuous variables. The dashed vertical lines indicate the threshold of |skewness| = 1 used to identify strongly skewed variables.

Figure 13 shows that several continuous measurements were strongly right-skewed. The most extreme case was `obesity_pct`, with a skewness of approximately 16.87, followed by ALT (7.28), AST (6.99), HDL (6.53), glucose (5.94), and CRP (5.41). Triglyceride was also strongly positively skewed (2.79), while GFR exhibited substantial negative skewness (−1.81). VFA showed a smaller but still notable positive skewness of approximately 1.06. In contrast, most of the remaining continuous measurements were relatively close to symmetric.

These results are consistent with the earlier outlier analysis. In particular, the extreme skewness of `obesity_pct` appears to be strongly influenced by the unusually large observation identified during data-quality assessment. Similarly, the long right tails observed for ALT, AST, glucose, CRP, and triglyceride indicate that a small number of participants have measurements considerably larger than the majority of the sample. These observations were not removed solely because they increased skewness, since extreme values may represent clinically plausible measurements.

4.3 Data Construction

The data-construction stage focused on producing a consistent analytical dataset while preserving the original source data. No additional derived clinical variables were introduced unless they had a clear analytical justification.

Three logical versions of the dataset were maintained:

- **Raw data:** the original dataset retained without modification;
- **Cleaned data:** the dataset after validation of field names, data types, categorical values, ranges, duplicates, and potential data-quality issues;

- **Model-ready data:** the final analytical dataset containing *Gallstone Status* and the predictor variables used by the Python modelling pipeline.

This separation ensures that preprocessing operations do not overwrite the source dataset and allows each transformation to be reproduced from the raw data through the Python scripts.

4.3.1 Data Partitioning

Iteration 4 uses five stratified 70% training / 30% test holdouts, with seeds 42, 7, 21, 84 and 123. Each split has 223 training and 96 test records. The disease-positive outcome is class 1 (source code 0). Saved source-row membership permits an audit of the disjoint training and test partitions.

Three-fold stratified cross-validation is performed exclusively within the 223 training records. Every inner fit learns its own imputers, 20-feature selector and continuous-variable transformations. The selected hyperparameters are then refitted on the complete training partition and evaluated once on its test partition at the fixed threshold 0.5. Stacking additionally generates training-only out-of-fold base probabilities for its meta-learner. Original numeric ordinal/binary category codes are retained without one-hot expansion.

The five holdouts reuse a cohort and therefore overlap; they measure sensitivity to split choice rather than five independent external validations. Test outcomes are descriptive evaluation evidence and cannot determine tuning settings or a validation-selected model.

4.4 Data Integration

Although the Gallstone dataset was originally obtained as a single public dataset, an explicit integration exercise was conducted to demonstrate how logically separate healthcare data sources can be combined into a unified analytical table.

The original dataset was preserved unchanged and divided into two logical CSV sources. The first file, `gall stone_demographic_clinical.csv`, contains 319 records and 9 fields describing demographic and clinical characteristics. The second file, `gallstone_biological_body_composition.csv`, contains the same 319 participants and 33 fields describing biological, laboratory, and body-composition measurements.

A unique `record_id`, ranging from 1 to 319, was included in both tables as the one-to-one integration key. Both tables also contained *Gallstone Status* so that target consistency could be verified before integration. However, *Gallstone Status* was not used as a merge key because it does not uniquely identify an individual participant.

Integration was implemented using the `pandas.merge()` operation with `record_id` as the matching field. The merge was configured as a one-to-one operation so that duplicate identifiers or unexpected many-to-one relationships could be detected explicitly.

Post-integration quality checks verified that all 319 participant records were retained and that no unmatched identifiers or additional missing values were introduced by the merge.

The duplicated *Gallstone Status* fields from the two logical data sources were compared before one copy was removed from the final analytical table. The `record_id` field was retained for traceability and quality control but excluded from the predictor set used for model development.

This integration exercise represents a simplified version of a realistic healthcare environment in which demographic and clinical data may be stored separately from laboratory or body-composition measurements. The Python workflow demonstrates how these sources can be integrated while maintaining record integrity and preventing inappropriate use of the target variable as an integration key.

4.5 Data Reformatting

The original Gallstone dataset was provided in Microsoft Excel format (`.xlsx`). The file was loaded programmatically using `pandas.read_excel()` and subsequently exported to comma-separated values (`.csv`) where appropriate for intermediate storage and reproducible processing.

CSV was selected as an intermediate format because it provides a simple, platform-independent representation of tabular data and avoids spreadsheet formatting that is irrelevant to the modelling process. The use of CSV was not a modelling requirement, however, because the Python pipeline can also operate directly on Excel or other structured tabular formats.

During reformatting, variable names were standardised and unnecessary spreadsheet-specific formatting was removed. The resulting fields were then classified according to their analytical meaning.

Continuous clinical and body-composition measurements, such as BMI, glucose, triglyceride, weight, and total body water, were represented as numerical variables.

Binary variables such as Coronary Artery Disease, Hypothyroidism, Hyperlipidemia, and Diabetes Mellitus were stored as categorical or binary indicators rather than being interpreted as continuous measurements.

Variables representing ordered clinical levels were treated as ordinal variables when their coding reflected an increasing degree of severity or disease burden.

The target variable, *Gallstone Status*, was represented internally as a binary label for model training while human-readable labels (*Gallstone Absent* and *Gallstone Present*) were used in tables and visualisations.

The final model-ready dataset therefore contained consistent field names, validated data types, clearly defined variable roles, and a reproducible schema that could be used directly by the subsequent Python-based feature engineering and modelling pipeline.

4.6 Reproducible Open-Source Preparation Pipeline

The complete preparation process was implemented as a script-based open-source workflow. At a high level, the pipeline consisted of:

1. loading the source data using `pandas`;
2. standardising variable names and data types;
3. validating missing values, duplicates, coding ranges, and observations;
4. performing exploratory outlier and distribution checks;
5. integrating logically separated datasets using `pandas.merge()`;
6. constructing the final predictor and target datasets;
7. creating stratified training/test holdouts and training-only CV folds using `scikit-learn`;
8. fitting any required preprocessing operations using the training subset only; and
9. passing the resulting datasets to the machine-learning modelling pipeline.

The use of an open-source Python stack provides several advantages for this project. Each operation is explicitly defined in code, preprocessing decisions can be version-controlled, experiments can be reproduced using fixed random seeds, and identical transformations can be consistently applied across multiple candidate models.

The resulting workflow therefore provides a transparent and reproducible transition from the original Gallstone dataset to the datasets used for model training, validation, and final evaluation.

Iteration 4 correction: preserve existing ordinal codes. The supplied predictors are already numerically encoded, including ordered codes such as 1, 2, 3, and 4 and binary indicators. These values are retained as provided: no one-hot expansion, category reindexing, or arbitrary replacement numbering is applied. Missing coded values are imputed by the training-partition mode; continuous measurements are median-imputed and scaled within training. The label mapping is separate: source gallstone status 0 remains model class 1 (gallstones present). Numeric category codes must reflect their documented order; storing a nominal category as a number does not itself establish ordinality. For logistic regression, a single ordinal coefficient imposes equal log-odds increments per code step, while trees can learn thresholds between levels. Historical fitted models and scores used the previous representation and do not validate this correction.

4.7 Data Reduction and Projection

4.7.1 Training-only Feature Selection

Both candidate selectors retain 20 original source fields. CART importance selection uses a decision tree with maximum depth 5 and minimum leaf size 4, retaining the 20 largest impurity importances. Recursive feature elimination (RFE) uses the same CART estimator, removes one field per step and refits until 20 remain. Imputation precedes selection, and the selected continuous fields are transformed after selection; ordinal category codes remain unchanged.

Both methods are supervised and therefore fitted only on training records. Every inner CV fit repeats imputation and selection, so its validation records cannot influence the retained fields. Stacking maintains a complete preparation pipeline for each base learner and refits selection inside its own out-of-fold construction. After tuning, the final pipeline refits selection using the complete 223-record training partition, before predicting the 96 held-out records.

Selected-feature CSVs include source names, selection flags and ranks, separately for each stacking base learner. Selection may vary across seeds and fits. Impurity importance can favour predictors with more available split points and does not establish a clinical effect. The four controlled preparation benchmarks in Section 8 evaluate selector and transformation changes while maintaining fixed model settings and test membership.

5 Data-Mining Method Selection

5.1 Learning Paradigm and Research Objective

The objective is to estimate gallstone presence from demographic, clinical, laboratory, and body-composition measurements. All 319 records have a known binary outcome, making **supervised learning** the appropriate primary paradigm: predictions can be compared with observed labels on held-out records. **Unsupervised learning** can reveal structure without those labels, but similarity between patients does not necessarily correspond to disease status. It therefore supports exploration rather than providing the principal answer to the research question.

5.2 Comparison of Data-Mining Methods

The learning paradigm must be narrowed to a method whose output matches the objective. Table 3 distinguishes this decision from the choice of a particular algorithm.

Table 3: Suitability of data-mining methods for the gallstone-prediction objective.

Method	Intended output	Suitability for this project
Classification	A known class and, where supported, its probability.	Directly matches labelled gallstone presence/absence and permits evaluation of discrimination, missed cases, and false alarms. Selected as the primary method.
Continuous-value prediction (regression)	A numerical quantity, such as a laboratory concentration or severity score.	The target is a category, not a continuous measurement. This would answer a different question. Classification is also predictive; the distinction here concerns the type of output.
Segmentation (clustering)	Groups of observations with similar predictor profiles.	Useful for exploratory subgroups, but groups need not correspond to gallstone status. Cluster membership alone is not a validated disease prediction.
Association discovery	Frequently co-occurring conditions or measurement patterns.	Useful for hypothesis generation, but support and confidence do not establish held-out discrimination or an acceptable false-negative rate. Outcome-directed rules would still require supervised classification evaluation.
Anomaly detection	Observations that differ from a reference pattern.	Supports review of unusual measurements, but gallstone cases comprise roughly half the sample and cannot be equated with anomalies.

Dimensionality reduction and feature selection are supporting preparation methods, not substitutes for predicting the labelled outcome. Likewise, forecasting future onset or modelling time to diagnosis would require longitudinal or event-time information not used by the current workflow. The available evidence supports classification of recorded gallstone status, not a demonstrated forecast of future disease.

5.3 Selected Method and Success Criteria

The selected method is **supervised binary classification with probability estimation**. It provides the required patient-level output and directly supports the project benchmarks of ROC–AUC at least 0.80 and gallstone-class sensitivity at least 0.80. Specificity, precision, F1, stability, and interpretability are needed alongside these measures: predicting every patient as positive could satisfy sensitivity while producing unacceptable false alarms.

The source coding is 0 = gallstones present and 1 = absent. The Python modelling target is recoded to 1 = present and 0 = absent, so $\mathrm{FNR} = \mathrm{FN}/(\mathrm{FN} + \mathrm{TP}) = 1 - \text{sensitivity}$ measures missed gallstone cases. These are analytical project benchmarks, not clinical deployment thresholds. The following section compares algorithms within this chosen method and explains how the implemented models are built and selected.

6 Algorithm Selection and Parameterisation

6.1 Exploratory Algorithm Analysis

Algorithm exploration uses three-fold training-only ROC–AUC, rather than final test performance. Logistic regression provides a compact additive reference; random forest tests nonlinear interactions with bagging; XGBoost and LightGBM test regularised boosting; stacking tests whether their out-of-fold probabilities provide complementary information. The training-prior dummy establishes chance discrimination. All candidate pipelines receive the same training partitions and inner folds. Imputation, feature selection and scaling are refitted within

every inner fold, including the base learners inside stacking.

Table 4: Algorithm exploration: training-only three-fold CV AUC, mean and sample SD across five seeds.

Model	Fixed CV AUC	Best-trial CV AUC
Logistic regression	0.809 ± 0.019	0.819 ± 0.025
Random forest	0.842 ± 0.021	0.852 ± 0.015
XGBoost	0.838 ± 0.016	0.854 ± 0.021
LightGBM	0.832 ± 0.024	0.848 ± 0.021
Stacking	0.845 ± 0.017	0.855 ± 0.015
Prior dummy	0.500 ± 0.000	Not tuned

Training-CV choices are seed 7: XGBoost (0.888), seed 21: Random forest (0.840), seed 42: Stacking (0.862), seed 84: Random forest (0.857), seed 123: Stacking (0.852). These are best-trial training scores, not test scores.

The strongest training-CV family is selected independently for each seed; test AUC does not replace this choice. Logistic regression remains an interpretable reference even if a more complex family has higher CV discrimination. Forest and boosting candidates are justified by plausible nonlinear clinical associations, but their added complexity needs measured training-CV support. Stacking is justified only where cross-validated evidence supports combining predictions; fitting more components alone is not evidence of improvement. Section 7 reports test outcomes separately, so this step documents the evidence used for selection rather than duplicating result tables.

6.2 Controlled Preparation and Candidate Settings

CART importance and CART-based recursive feature elimination (RFE) each retain 20 original fields. Both are tested with Yeo–Johnson followed by standardization, and with standardization alone. Ordinal category codes pass through unchanged. These four fixed-parameter benchmarks isolate selector and transformation changes; the predeclared RFE/standardization configuration is used for the tuning comparison, without selecting it from test results.

6.3 Training-only Optuna Parameter Search

Each model receives 20 Optuna TPE trials for each of five seeds, scored by mean three-fold training ROC–AUC. Standalone logistic regression searches $C \in [0.01, 100]$ on a logarithmic scale. Random forest searches tree count 40–160 in steps of 20, maximum depth 2–8, minimum leaf size 2–10 and maximum feature fraction (square root, 0.5, 1.0). XGBoost searches tree count 40–160, learning rate 0.01–0.15 (log), depth 1–4, row and column sampling 0.6–1.0, and L2 regularisation 0.1–10 (log). LightGBM searches tree count 40–160, learning rate 0.01–0.15 (log), leaves 3–15, minimum child records 10–40, column sampling 0.6–1.0 and L2 regularisation 0.1–10 (log).

Stacking searches the twelve settings of its layer-1 XGBoost and LightGBM pipelines jointly. Its base logistic regression and meta logistic regression remain fixed at $C = 1$; its forest and three-fold stacking configuration remain fixed. RFE retains 20 fields with step 1 and a CART selector of depth 5 and minimum leaf size 4. These selector controls are compared through the controlled preprocessing benchmarks rather than included in the classifier search. The fixed search budget is reproducible but cannot establish exhaustive optimisation. Per-model trial CSVs, best-parameter JSONs and fitted pipelines document all tested values, including unsuccessful improvement attempts.

7 Model Execution and Statistical Results

7.1 Repeated Holdout Execution

Five seeds (42, 7, 21, 84, 123) each produce a 223-record training partition and a 96-record test partition. Four preparation variants are benchmarked with fixed settings for five predictive families and the training-prior dummy. The predeclared RFE/standardization variant additionally receives training-only Optuna searches for all five predictive families. The dummy has no tunable hyperparameters. No test record enters an inner CV fit or parameter search.

Table 5: RFE20/standardization test results: mean and sample SD across five holdouts. Highest column means are bold, including the prior dummy; exact ties share highlighting.

Model	AUC	Sensitivity	Specificity	Accuracy
Prior dummy (benchmark)	0.500 ± 0.000	**1.000** ± 0.000	0.000 ± 0.000	0.500 ± 0.000
LightGBM (benchmark)	0.799 ± 0.027	0.775 ± 0.063	0.679 ± 0.035	0.727 ± 0.028
Logistic regression (benchmark)	0.832 ± 0.042	0.817 ± 0.063	0.688 ± 0.081	0.752 ± 0.046
Random forest (benchmark)	0.808 ± 0.047	0.783 ± 0.041	0.700 ± 0.075	0.742 ± 0.041
Stacking (benchmark)	0.824 ± 0.029	0.787 ± 0.048	0.708 ± 0.074	0.748 ± 0.017
XGBoost (benchmark)	0.801 ± 0.030	0.796 ± 0.073	0.688 ± 0.042	0.742 ± 0.032
LightGBM (tuned)	0.793 ± 0.015	0.779 ± 0.079	0.667 ± 0.044	0.723 ± 0.030
Logistic regression (tuned)	**0.843** ± 0.037	0.808 ± 0.071	0.700 ± 0.079	0.754 ± 0.041
Random forest (tuned)	0.803 ± 0.032	0.804 ± 0.043	0.658 ± 0.068	0.731 ± 0.027
Stacking (tuned)	0.824 ± 0.038	0.812 ± 0.042	**0.713** ± 0.063	**0.762** ± 0.039
XGBoost (tuned)	0.793 ± 0.033	0.825 ± 0.062	0.667 ± 0.047	0.746 ± 0.032

Both project targets were met in LightGBM: 2/5 splits, Logistic regression: 2/5 splits, Random forest: 2/5 splits, Stacking: 3/5 splits, XGBoost: 3/5 splits. Full metric distributions and all four benchmark variants are saved in the result CSVs.

Metrics use disease-positive class 1 and threshold 0.5. Outputs include ROC–AUC, average precision, accuracy, balanced accuracy, precision, sensitivity, specificity, F1, false-negative rate, log loss, confusion counts and fit/prediction time. Mean and sample standard deviation describe five split-specific results; these overlapping splits do not supply independent replication. Each model folder contains predictions, selected fields, fitted pipeline and parameter evidence. Benchmark flags test the project targets AUC ≥ 0.80 and sensitivity ≥ 0.80.

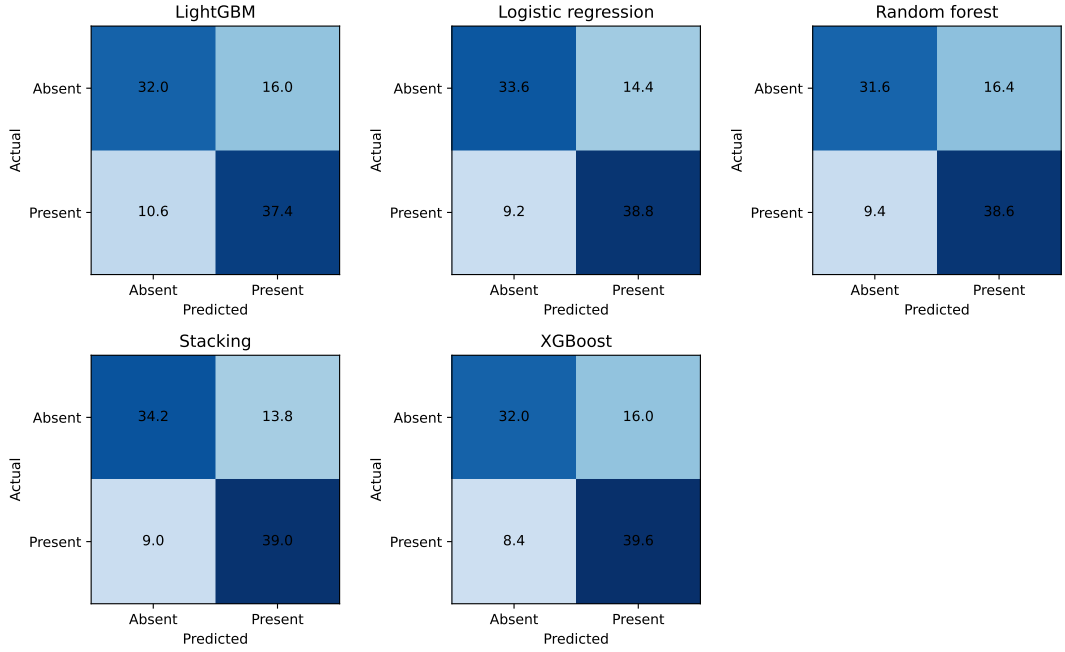


Figure 14: Average test confusion counts over five seeds for each tuned predictive family. Fractional counts arise from averaging 96-record test sets; they do not represent a single pooled independent cohort.

8 Interpretation, Evaluation and CRISP-DM Iteration

8.1 Controlled Iterations and Evaluation

Iteration A benchmarks CART selection with Yeo–Johnson plus standardization. Iteration B changes only continuous-variable scaling to standardization. Iteration C replaces CART importance selection with RFE, retaining 20 fields and Yeo–Johnson plus standardization. Iteration D uses RFE with standardization alone. Model settings, test membership and seeds are held constant across these comparisons. This separates preparation effects from changes in model parameters.

Iteration E applies 20-trial training-only Optuna searches to all five predictive families in the predeclared RFE/standardization configuration. It includes joint tuning of the stacking layer-1 boosting learners while retaining both logistic components unchanged. Matched benchmark and tuned test scores evaluate the effect of the tuning procedure; the test scores never enter its objective.

Table 6: Controlled fixed-model preparation comparisons: mean test AUC over five matched seeds.

Model	CART+YJ	CART+Std	RFE+YJ	RFE+Std
Logistic regression	0.861	0.836	0.864	0.832
Random forest	0.810	0.811	0.808	0.808
XGBoost	0.809	0.809	0.801	0.801
LightGBM	0.805	0.814	0.794	0.799
Stacking	0.845	0.839	0.837	0.824

Table 7: Matched tuning effects on RFE/standardization: tuned minus fixed benchmark.

Model	Mean AUC change	Mean sensitivity change	AUC gains
LightGBM	-0.006	+0.004	1/5
Logistic regression	+0.011	-0.008	4/5
Random forest	-0.005	+0.021	2/5
Stacking	-0.000	+0.025	2/5
XGBoost	-0.008	+0.029	1/5

Improvement is assessed per family and per seed rather than assumed from the existence of a larger search. A positive mean AUC difference can coexist with reduced sensitivity at threshold 0.5. Consequently, the report checks both screening targets, reports false negatives and retains unsuccessful changes as evidence. Search scores are optimistic estimates after selecting the best of 20 trials; the untouched test partition supplies the corresponding internal evaluation.

The measured tuning changes are mixed: logistic regression gains approximately 0.011 mean test AUC while sensitivity decreases by 0.008. Random forest, XGBoost and LightGBM have mean AUC changes of approximately -0.005, -0.008 and -0.006, respectively, while sensitivity increases by approximately 0.021, 0.029 and 0.004. Stacking changes AUC by approximately -0.0004 and gains 0.025 sensitivity. These matched outcomes do not justify claiming that a larger parameter search uniformly enhances discrimination; it changes the sensitivity/discrimination trade-off under the fixed threshold.

8.2 Interpretation and Next Iteration

Training-CV family choices are reported in Section 6.1; detailed test metrics appear once in Section 7. Section 8 compares controlled changes rather than restating those tables. The current evidence supports internal algorithm comparison only. The overlapping five-seed test sets, small single-source cohort, fixed decision threshold and finite parameter budget limit generalisation. Further work would predefine clinically appropriate threshold/calibration procedures inside training, assess feature-selection stability and validate the locked procedure on an external cohort.

9 Limitations

The experiments use a single 319-record cohort. Five holdout splits overlap and cannot be treated as independent clinical replications. Training-only cross-validation protects test separation, but selecting the best trial creates optimism in the reported CV score. Twenty trials explore the documented ranges without proving a global optimum. Threshold 0.5 is fixed; screening suitability requires external assessment of sensitivity, calibration and clinical consequences.

Ordinal categories retain their supplied numeric codes. This preserves the dataset representation but imposes ordering for linear models; it remains a modelling assumption. The principal five-family implementation uses scikit-learn, XGBoost and LightGBM. The optional native Spark workflow uses Spark's own logistic regression, random forest and gradient-boosted trees and is a distinct supplementary implementation; its results are not attributed to these Python-library models.

10 Acknowledgement

I acknowledge that this submission is my own original work and has been prepared in accordance with the University of Auckland's policies on academic integrity and copyright. Further information is available from the University of Auckland Academic Integrity and Copyright guidelines.

I also confirm that I have the appropriate permission to use the data included in this project. Where data are owned by an organisation and are not publicly available, permission must be obtained from the relevant rights holder, including permission to upload the data to Canvas. The University of Auckland accepts no responsibility for any misuse of data by the student.

Use of Artificial Intelligence

Generative artificial intelligence tools were used to support language editing, sentence restructuring, LaTeX formatting, and the organisation of the report. All project decisions, data analysis, interpretation of results, and final content were reviewed and completed by the author. No confidential, personal, or restricted data were shared with an artificial intelligence tool.

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A Instructions for Running the Submitted Workflow

`iteration4_BDAS/.venv` links to the existing `../Experimental/.venv`; no dependencies were reinstalled. From the Iteration 4 directory run:

```
.venv/bin/python run_workflow.py --trials 20
.venv/bin/python scripts/build_holdout_report.py
cd Report
latexmk -pdf -interaction=nonstopmode main.tex
```

The main notebook `run_workflow.ipynb` invokes the same workflow. Results are saved under `results/holdout/seed_<seed>/variant/benchmark_or_tuned/model/`. Identical configurations resume completed model evaluations. When the default saved configuration differs, the workflow automatically chooses a separate configuration-specific directory under `results/holdout_runs/`; identical reruns resume there. Explicit incompatible directories are rejected with the changed fields listed. The notebook uses `metrics.attrs['output_dir']` to locate summaries and generate figures for its own run. The saved configuration records package versions and data signature. The shared environment link is local and must be replaced by an available environment on another machine; the notebook checks dependencies without installing packages.